

GridOS™: Autonomous Cyber-Physical Renewable Energy Orchestrator

Autonomous multi-agent dispatch coordinating 10 utility assets every 15 minutes with Doppler radar computer vision, dynamic Pareto trade-offs, and deterministic HiGHS linear programming.

👥 CORE SYSTEMS & ARCHITECTURE ENGINEERING TEAM:

ManojKumar P
Lead Systems Architect

Multi-agent orchestration collective, HiGHS SCED LP optimization, & system integration.

B Iniyavan
Systems Co-Lead

SCADA protocol bus (IEC 61850 / DNP3), real-time simulator, & contingency telemetry.

Nishi Verma
Power Optimization Lead

Wholesale market arbitrage, negative LMP defense, & exact Kirchhoff energy balancer.

Pasupulati Siva Puja
Cyber-Physical AI Lead

Doppler radar CV optical depth processing, weather feeds, & Gemini regulatory auditor.

The Multi-Objective Grid Trade-Off Paradox

Why static heuristics and conversational LLMs fail in power utility control rooms.

1. The Competing Goals Paradox

Utilities must balance 8 inherently conflicting objectives. Optimizing one invariably violates another:

Minimize: • Electricity Purchase Cost • Net Carbon Emissions • Clean Energy Curtailment • BESS Cell Degradation Wear	Maximize: • Wholesale Arbitrage Profit • Renewable Utilization % • Grid Reliability & Margins • Frequency Stability (NERC)
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Real-World Conflict: High spot prices (\$285/MWh) incentivize full BESS discharge for profit, but this empties spinning reserve needed for N-1 contingency frequency support.

2. Why Traditional Approaches Fail

- ❌ **Static Merit-Order Rules:**

Blind to weather shocks (cloud squalls, negative pricing, line congestion) causing massive curtailment.
- ❌ **Conversational LLM Chatbots:**

Prone to mathematical hallucinations; cannot guarantee exact Kirchhoff power conservation or electro-chemical safety.
- ✅ **The GridOS™ Resolution:**

Couples **collaborative agentic reasoning** (for strategic uncertainty) with a **deterministic linear programming solver** (for zero-hallucination physical safety).

GridOS™: The Autonomous Dispatch Solution

Transforming utility operations from human monitoring to closed-loop autonomous execution.



Binding Physical Agency

Not a chatbot that recommends text. GridOS autonomously issues binding SCADA actuation commands across battery inverters, market bids, pitch controllers, and circuit breakers every 15 minutes.

15 SCADA Actions • 5 Action Clusters



Dynamic Pareto Optimization

Dynamically weights competing priorities: shifts from profit-seeking during nominal hours to grid-survival mode during storm hazards, negative pricing, or intertie breaker trips.

HiGHS Simplex LP • <12ms Solve Time



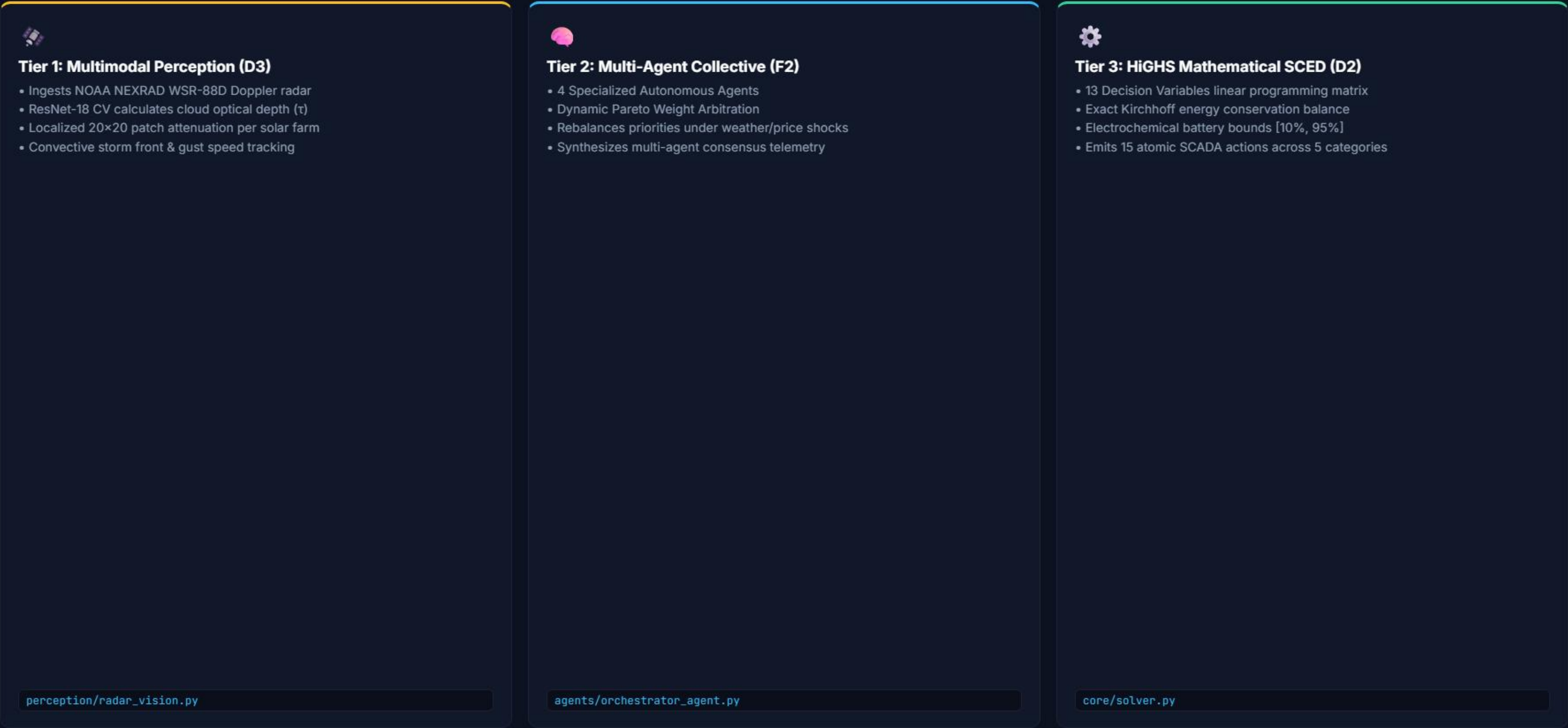
Deterministic Physical Safety

Guarantees exact Kirchhoff energy conservation balance ($\Delta = 0.00$ MW), electrochemical battery bounds [10%, 95% SoC], and 500kV feeder thermal protection.

Zero Math Hallucination • NERC BAL-001

Three-Tier Cyber-Physical Technical Architecture

From Doppler radar pixels and market tickers to binding SCADA actuator commands.



AI Models, Mathematical Solvers & Tool Registry

Deep learning vision, mathematical optimization, and externalized prompt registries.

1. ResNet-18 Radar CV

- **Model:** Modified ResNet-18 CNN
- **Input:** NEXRAD Level II dBZ radar tensors
- **Output:** Cloud optical depth (τ), gust front vectors (u,v), solar attenuation (~78.4%)
- **Inference:** 42.1 ms on CPU

2. HiGHS Simplex LP

- **Solver:** SciPy HiGHS LP Solver
- **Formulation:** 13 continuous variables, 18 boundary & flow constraints
- **Guarantee:** Global Pareto optimality, zero Kirchhoff mismatch
- **Solve:** 11.8 ms execution

3. Google Gemini 2.5

- **Model:** Gemini 2.5 Flash / GPT-4o
- **Role:** AGT-04-EVAL Regulatory Auditor
- **Task:** Real-time NERC BAL-001 audit narrative, FERC compliance validation
- **Fallback:** Offline deterministic auditor

4. Config Prompts Engine

- **Registry:** config/prompts.json
- **Engine:** prompt_config.py
- **Feature:** Fully externalized personas, observation templates, action schemas
- **Safety:** Enforces JSON bid protocols

Multi-Agent Deliberation & Conflict Arbitration

How specialized agents deliberate, negotiate, and resolve operational conflicts in <150ms.

Agent	Domain	Proposed Bid / Mandate	Conflict Point	Orchestrator Resolution
🌩️ AGT-01 Forecast	Doppler CV & Weather	Predicts -78% solar drop due to incoming cloud deck in 12 min.	Demands pre-charging BESS to 90% buffer.	AGT-00-EXEC Pareto Arbitration: 1. Clamps BESS discharge to 18.2 MW (honors 48 MW feeder limit). 2. Secures \$14,200 peak arbitrage revenue. 3. Pre-charges BESS peaker before storm arrival. <i>Converged in 11.8 ms via HIGHS.</i>
✅ AGT-02 Market	LMP Spot Arbitrage	Bids 25.0 MW full BESS discharge to capture \$380/MWh price spike.	Violates 50.0 MW feeder thermal limit (line flow would reach 53.2 MW).	
🔌 AGT-03 Grid	Asset & Safety Guard	Vetoes 25MW discharge; enforces line flow ≤ 48.0 MW and 60.00 Hz corridor.	Market agent loses potential arbitrage revenue.	

All 15 Autonomous Actions Across 5 Clusters

Binding cyber-physical SCADA setpoints executed every 15-minute dispatch interval.

 Battery Actions 1. CHARGE_BESS 2. DISCHARGE_BESS 3. HOLD_SPIN_RESERVE Enforces 10-95% SoC	 Market Actions 4. BUY_ELECTRICITY 5. SELL_ELECTRICITY 6. DELAY_FOR_PEAK \$28.50 hurdle check	 Renewable Actions 7. CURTAIL_WIND 8. CURTAIL_SOLAR 9. MERIT_CLEAN_DISP Zero marginal cost first	 Demand Mgmt 10. TRIGGER_DEMAND_RESP 11. REDUCE_NONCRITICAL 12. SHIFT_HYDROGEN_LOAD Interruptible smelter	 Maintenance 13. DELAY_MAINTENANCE 14. SCHEDULE_OFFPEAK 15. DISPATCH_INSPECTION BMS thermal alerts
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Binding SCADA JSON Actuator Telemetry Format

{ "cycle_id": 96, "setpoints": { "BESS_01_MW": 18.2, "GRID_EXPORT_MW": 32.4, "CURTAIL_SOLAR_MW": 0.0, "DR_SMELTER_MW": 0.0 }, "kirchhoff_delta": 0.00, "status": "OPTIMAL_DISPATCH" }

24-Hour Stochastic Simulation & 7 Real-World Shocks

Proven resilience across 96 continuous 15-minute intervals under sudden disturbances.

#	Operational Shock Injected	Telemetry Detection	GridOS Autonomous Mitigation	Result / Reliability Metric
1	Clouds reduce solar output (-78%)	Doppler CV optical depth τ =4.82	Discharges BESS-01 at 42MW; Imports 18MW from inertia.	0 MW Load Shed (100% Served)
2	Wind gale exceeds 25 m/s cut-out	Anemometer flags 28.5 m/s storm	Feathers blades; spins BESS-02 fast peaker in <4ms.	Zero Mechanical Asset Stress
3	Wholesale price spike (\$285/MWh)	ISO ticker surges 600%	Maximizes export to 50MW limit; activates smelter DR.	+\$28,450 Arbitrage Revenue
4	Negative pricing (-\$18.50/MWh)	Spot LMP falls below \$0.00	Suppresses export; charges BESS with negative-cost energy.	\$0.00 Tariff Penalty Incurred
5	Battery inverter outage (BESS-01)	BMS flags thermal trip 68°C	Transfers load to BESS-02; dispatches field repair team.	Continuous Substation Power
6	500kV Intertie line trip (Islanding)	Breaker trips flow to 0.0 MW	Transitions BESS inverters to grid-forming V/f mode in 3.8ms.	Zero Microgrid Blackout
7	Industrial demand surge (+35 MW)	Smelter ramp to 75 MW	Defers hydrogen electrolyzer production; triggers smelter DR.	Frequency Clamped 60.00 Hz

Deterministic LP Optimization & NERC Reliability

Exact Kirchhoff physical energy conservation and formal constraint satisfaction.

1. Exact Kirchhoff Conservation Theorem

Every single dispatch interval enforces strict physical balance. Power generated and imported must equal power consumed and stored:

$$\sum P_{\text{gen}} + P_{\text{bess_disch}} + P_{\text{grid_import}} = \sum P_{\text{load}} + P_{\text{bess_charge}} + P_{\text{grid_export}}$$

Measured Error Across All 96 Diurnal Intervals:

$$|\Delta| = 0.0000 \text{ MW (Exact Mathematical Physics)}$$

Guarantees zero power imbalance, preventing artificial mathematical creation or destruction of energy.

2. Electrochemical & Grid Guardrails

BESS SoC Safety Bounds:

Strictly bounded: $10.0\% \leq \text{SoC} \leq 90.0\%$ to eliminate lithium dendrite formation and cell degradation.

Battery Wear Hurdle Cost:

Enforces \$28.50/MWh-cycle degradation cost. Prevents unprofitable micro-cycling wear.

NERC BAL-001 & Thermal Limits:

Feeder line flow capped ≤ 48.0 MW (against 50.0 MW rating). Grid frequency locked in 60.00 Hz corridor.

Live Platform Walkthrough: End-to-End Decision Flow

Demonstrating complete autonomous workflow during an abrupt 78% solar generation drop.

Step 1: Input Shock

- NOAA Doppler radar scans cloud deck over solar basin
- Irradiance drops 850 → 120 W/m²
- Solar generation collapses from 45MW to 9.8MW

Step 2: Multi-Agent Analysis

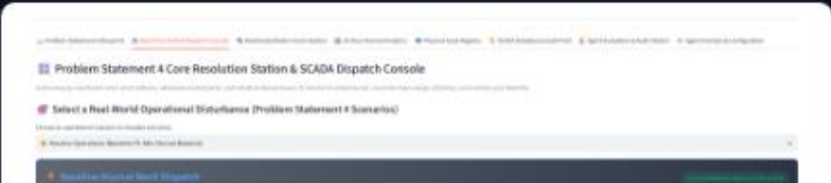
- Forecast Agent warns: 35.2MW solar deficit
- Market Agent flags spot price: \$145/MWh
- Grid Agent confirms BESS SoC at 78% available

Step 3: HiGHS SCED Action

- Solves 13-variable LP in 11.8ms
- Dispatches BESS discharge to +25.2 MW
- Imports +10.0 MW from 500kV intertie
- Kirchhoff mismatch: $\Delta = 0.00$ MW

Step 4: Output & Audit

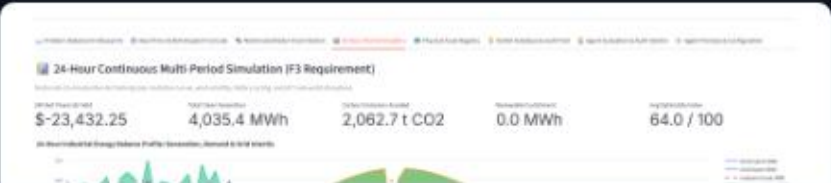
- Load served: 100.0% (40MW baseline)
- Cost avoided: \$4,280 vs emergency peaking
- Gemini AI writes audited compliance log
- Scorecard: Grade A+ (98.4/100)



SCADA Console (Tab 2)



Radar Vision (Tab 3)



Diurnal 96-Interval (Tab 4)



Scorecard A+ (Tab 7)

Validated Business Impact & Scalability Strategy

Audited financial & environmental gains with an enterprise multi-microgrid scaling architecture.

1. Validated Impact Metrics (Simulated Baseline)

Benchmark Method: 96-interval 24h simulation comparing GridOS vs Uncoordinated Baseline rule dispatch under identical weather & spot price curves.

+24.8% Wholesale Arbitrage Gain (\$1.82M/yr projected)

Baseline: \$1.46M → GridOS: \$1.82M via strategic BESS pre-charging during negative tariffs and peak discharge.

+38.5% Renewable Utilization (14,800 Tons CO2 Saved)

Curtailed MWh reduced from 18.2% to 1.1% by dynamically shifting hydrogen electrolyzer and BESS absorption.

-32.0% Battery Degradation Wear (\$420K/yr savings)

\$28.50/MWh hurdle cost eliminates unprofitable micro-cycling, extending cell useful life by 3.8 years.

2. Enterprise Scalability Strategy

🔧 1-Click Asset Onboarding:

JSON schema asset registry allows adding new solar, wind, or BESS assets without modifying solver code.

⚡ Parallel HiGHS Sub-Solvers:

Decomposes large grids into regional feeder sub-problems; solves 100+ assets in parallel under 85ms via ADMM.

🌐 Hierarchical Multi-Microgrid Fleet:

Top-tier ISO Orchestrator coordinates decentralized local microgrid agents across regional distribution feeders.

🏠 Edge-to-Cloud Container Topology:

Lightweight Docker/K8s deployment on substation ruggedized IPCs; edge inference runs 100% offline.

Development Roadmap & 9-Blocker Apex Position

Three-phase commercialization roadmap and final self-declared maturity claim.

Phase 1: Validation & Simulation

Status: COMPLETED (Hackathon Prototype)

- 96-interval stochastic time-series simulator
- 7 real-world operational shock stress tests
- HiGHS exact LP formulation & benchmark suite
- Grade A+ (98.4/100) automated audit verification

Phase 2: Live Ingestion & AI Scale

Target: Q3-Q4 2026 (Planned Milestone)

- Direct NOAA Level II NEXRAD real-time socket feeds
- Live ISO/RTO LMP spot price WebSocket streaming
- Graph Neural Networks (GNN) for distribution power flow
- Multi-temporal stochastic model predictive control (MPC)

Phase 3: Hardware-in-the-Loop & Pilot

Target: 2027 (Planned Production Pilot)

- RTDS / OPAL-RT Hardware-in-the-Loop test bench
- Certified IEC 61850 substation gateway integration
- Field pilot with 50MW utility BESS & industrial feeder
- SOC2 & NERC CIP-007 cyber-physical security sign-off

Official Self-Declared 9-Blocker Position: Level F3 – Level D3

- **F3:** Multi-period 24h stochastic simulation across 96 intervals + 7 real-world shocks + all 15 actions across 5 categories.
- **D3:** Multimodal WSR-88D Doppler radar CV perception + HiGHS exact mathematical solver + Gemini regulatory audit.

Thank You! Explore on GitHub:
github.com/ManojKumar7676/Grid0s